DevFest ગાંધીનગર

FOR THE FIRST TIME IN GUJARAT

Metro Hackathon

PARTICIPANT BRIEF · PROBLEM STATEMENTS



Two hours. One metro ride. Real problems from the people who run the city’s transport.

{ About the Hackathon }

This is a two-hour hackathon held on board the Gujarat Metro. Every problem in this brief comes from the day-to-day challenges faced by the people who run the metro and the city's public transport. Your task is to pick one problem and build a working prototype before the ride ends.

The problems fall into two tracks:

- **Metro Operations:** challenges faced by the metro operator in running stations, trains and assets.
- **City Public Transport:** challenges in connecting the metro with buses, roads and the wider city.

{ Things to Know Before You Start }

- **Expect connectivity drops.** Network coverage on the train is patchy, especially in underground sections. Download any libraries, models and reference material you need before boarding, and plan your work so you can push code or sync data when the train is at a station.
- **Offline problems are tested offline.** Problems marked "Offline operation: Required" must keep working with no network. Judges may switch on airplane mode during your demo.
- **Bring your own data.** No datasets will be provided. Use publicly available data, or create realistic sample data that fits your chosen problem, and be ready to explain your assumptions.
- **Scope for two hours.** A focused, working demo of one core flow beats an ambitious idea that doesn't run.

{ Judging Criteria }

- **Relevance to the problem:** Does it address the stated problem in a way an operator could actually use?
- **Working prototype:** Does the core flow work end to end in the demo?
- **Resilience:** Does it hold up with poor or no network, and does it degrade gracefully?
- **Technical depth:** Is the technology, including any AI, doing meaningful work rather than decoration?
- **Feasibility and impact:** Could this realistically be piloted by the metro or the city?

{ Problem Statements at a Glance }

ID	Title	Track	Offline
PS-01	Winning Back the Empty Seats	Metro Operations	Not required
PS-02	Fix It Before It Breaks	Metro Operations	Not required
PS-03	The Technician in the Tunnel	Metro Operations	**Required**
PS-04	No Day Is an Average Day	Metro Operations	Not required
PS-05	The Stadium Surge	Metro Operations	Not required
PS-06	Footfall to Revenue	Metro Operations	Not required
PS-07	Every Complaint Heard	Metro Operations	**Required**
PS-08	One City, One Journey	City Public Transport	Not required
PS-09	Clear the Exit	City Public Transport	**Required**
PS-10	Diversions Without Disruption	City Public Transport	Not required
PS-11	Beat the Heat	City Public Transport	Not required
PS-12	Monsoon Mode	City Public Transport	Not required
PS-13	The Two-Minute Morning Brief	City Public Transport	Not required

PS-01 · METRO OPERATIONS

{ Winning Back the Empty Seats }

PROBLEM STATEMENT

Metro trains run below capacity while the roads beneath them stay congested. Identify which commuters the metro is failing to attract, and why.

DESCRIPTION

The metro corridor can carry far more passengers than it currently does, and every empty seat is lost fare revenue. Potential riders include two-wheeler users, BRTS and city bus users, and office-goers who find the last mile from the station too difficult. Operators currently lack a clear view of which commuter segments offer the highest conversion potential and what would make them switch.

EXPECTED DELIVERABLE

Using station-wise, hour-wise ridership data and commuter survey data, identify the three commuter segments with the highest conversion potential. For each, propose a targeted, data-backed intervention (e.g. a fare product, a feeder route, an employer tie-up).

Offline operation	Not required

PS-02 · METRO OPERATIONS

{ Fix It Before It Breaks }

PROBLEM STATEMENT

Ticket gates, vending machines, lifts and escalators fail without warning, and maintenance is reactive.

DESCRIPTION

Asset faults are recorded in registers and spreadsheets, but the data is rarely used to anticipate failures. A failed ticket gate creates visible queues at the station before the control room is even informed. Operators need a way to predict which assets are likely to fail and plan maintenance accordingly.

EXPECTED DELIVERABLE

From historical asset fault logs, flag the assets most likely to fail in the coming week and generate a prioritized maintenance roster for the night shift, viewable by technicians.

Offline operation	Not required

PS-03 · METRO OPERATIONS

{ The Technician in the Tunnel }

PROBLEM STATEMENT

Maintenance staff often work in underground sections and equipment rooms with no network, while the manuals they need are long PDF documents.

DESCRIPTION

When a technician faces an unfamiliar fault, finding the right troubleshooting procedure in a lengthy technical manual is slow, and cloud-based tools are unusable without connectivity. The solution must work fully on-device.

EXPECTED DELIVERABLE

An offline assistant where a technician describes or photographs a fault and receives the relevant troubleshooting steps from a technical manual stored on the device, in Gujarati or English. Teams are encouraged to test it live on the train.

Offline operation	**Required**

PS-04 · METRO OPERATIONS

{ No Day Is an Average Day }

PROBLEM STATEMENT

Train frequency is planned for a typical day, but exams, festivals, rain and cricket matches shift demand significantly.

DESCRIPTION

Headways (the time gap between trains) are planned around average peak patterns. In reality, demand varies with events, weather and holidays, causing overcrowding on some days and empty trains on others.

EXPECTED DELIVERABLE

A recommender that takes the next day's calendar (events, weather forecast, holidays) and suggests headway changes by time band, with a plain-language justification for each change that a control room operator can act on.

Offline operation	Not required

PS-05 · METRO OPERATIONS

{ The Stadium Surge }

PROBLEM STATEMENT

Large events send thousands of people to one or two stations within minutes, creating serious crowd-safety risks.

DESCRIPTION

Cricket matches, Navratri nights and large riverfront events produce sudden surges at nearby stations. Operators know the event dates but lack a ready, actionable crowd management plan.

EXPECTED DELIVERABLE

Given an event's details (venue, expected attendance, end time), generate a minute-by-minute crowd plan: which gates to open, where to hold passengers, when to add trains, plus announcement scripts in Gujarati, Hindi and English.

Offline operation	Not required

PS-06 · METRO OPERATIONS

{ Footfall to Revenue }

PROBLEM STATEMENT

Retail spaces and advertising panels at stations remain vacant because their value cannot be demonstrated.

DESCRIPTION

Non-fare revenue is important to a metro's finances, but operators struggle to justify pricing to advertisers and retailers without credible evidence of footfall and audience profile.

EXPECTED DELIVERABLE

Convert station footfall data into a data-backed rate card and generate a one-page sales pitch for each vacant retail or advertising spot.

Offline operation	Not required

PS-07 · METRO OPERATIONS

{ Every Complaint Heard }

PROBLEM STATEMENT

Passenger complaints arrive through multiple channels and languages, and urgent ones get buried.

DESCRIPTION

Grievances come through the app, social media, email and paper forms, written in Gujarati, Hindi, English and mixed "Gujlish". Manual sorting by urgency and department takes hours, and a station controller may need to triage without network access.

EXPECTED DELIVERABLE

A tool that sorts a mixed batch of complaints into department queues (safety, cleanliness, fare, lifts, staff behaviour) with urgency scores and draft replies in the complainant’s language. It must work on-device and sync when connectivity returns.

Offline operation	**Required**

PS-08 · CITY PUBLIC TRANSPORT

{ One City, One Journey }

PROBLEM STATEMENT

Metro, BRTS and city buses are run by different operators with separate tickets, and transfers between them are confusing.

DESCRIPTION

A commuter who has to combine metro, BRTS and AMTS legs gets no single view of the journey, its total cost or the transfer walks involved. Many give up and use private vehicles instead.

EXPECTED DELIVERABLE

A journey planner prototype that combines metro, BRTS and AMTS legs for a sample trip, showing one fare estimate and transfer walking times. Bonus: propose a fare-splitting logic between the operators.

Offline operation	Not required

PS-09 · CITY PUBLIC TRANSPORT

{ Clear the Exit }

PROBLEM STATEMENT

Metro station exits are choked by clustered auto-rickshaws, illegal parking and encroachments.

DESCRIPTION

Congestion at station exits slows traffic and makes access unsafe for passengers. Enforcement teams lack a clear, prioritized view of where problems occur and how often. Field staff often report from spots with weak network.

EXPECTED DELIVERABLE

A field app where staff photograph a station exit; the app classifies the issue (auto cluster, illegal parking, encroachment, obstruction), logs it offline, and produces a daily hotspot map for enforcement teams.

Offline operation	**Required**

PS-10 · CITY PUBLIC TRANSPORT

{ Diversions Without Disruption }

PROBLEM STATEMENT

Construction-related traffic diversions change frequently, and citizens learn about them only when they are stuck.

DESCRIPTION

Ongoing metro extension and road works cause regular diversions. Advisories are often late, unclear, or issued in a single language, and affected bus routes are not communicated.

EXPECTED DELIVERABLE

Take a construction schedule as input and automatically generate diversion advisories: a map, multilingual text, WhatsApp-ready messages, and a list of affected bus routes.

Offline operation	Not required

PS-11 · CITY PUBLIC TRANSPORT

{ Beat the Heat }

PROBLEM STATEMENT

Extreme summer heat discourages people from walking to and from metro stations.

DESCRIPTION

With summer temperatures in Ahmedabad crossing 45°C, even a short walk to a station is a deterrent. The city needs to know which walking routes lack shade and where investment in shelters or tree cover would help most.

EXPECTED DELIVERABLE

A heat-aware last-mile planner that suggests shaded walking routes to stations, flags unshaded stretches, and ranks locations where the city should add shelters or tree cover first.

Offline operation	Not required

PS-12 · CITY PUBLIC TRANSPORT

{ Monsoon Mode }

PROBLEM STATEMENT

Heavy rain floods station entrances, approach roads and parking areas, and the response is usually reactive.

DESCRIPTION

Waterlogging during the monsoon disrupts access to stations. Operators and civic teams need advance, station-specific guidance before a heavy rainfall event rather than after.

EXPECTED DELIVERABLE

Using a rainfall forecast along with drainage and elevation data, produce a 24-hour readiness plan for each station: which entrances to close, where to deploy pumps, and what to communicate to commuters.

Offline operation	Not required

PS-13 · CITY PUBLIC TRANSPORT

{ The Two-Minute Morning Brief }

PROBLEM STATEMENT

Daily management reports are compiled manually from several systems, taking hours to prepare.

DESCRIPTION

Daily MIS reports for the state government and ministries require combining data from metro, BRTS and AMTS systems. The effort is high, and important exceptions are easy to miss.

EXPECTED DELIVERABLE

Take the raw daily data from all three systems and generate a one-page brief with exceptions highlighted and likely causes suggested (e.g. "Ridership at Station X down 12%; likely cause: gate outage plus rain").

Offline operation	Not required